

3.2.3 Solving equations and inequalities

A17

Basic foundation content	Additional foundation content	Higher content only
solve linear equations in one unknown algebraically find approximate solutions using a graph	including those with the unknown on both sides of the equation	

Notes: including use of brackets.

A18

Basic foundation content	Additional foundation content	Higher content only
	solve quadratic equations algebraically by factorising	including those that require rearrangement including completing the square and by using the quadratic formula
	find approximate solutions using a graph	

Notes: see also [A11](#)

A19

Basic foundation content	Additional foundation content	Higher content only
	solve two simultaneous equations in two variables (linear/linear) algebraically find approximate solutions using a graph	including linear/quadratic

A20

Basic foundation content	Additional foundation content	Higher content only
		find approximate solutions to equations numerically using iteration

Notes: including the use of suffix notation in recursive formulae.

A21

Basic foundation content	Additional foundation content	Higher content only
	translate simple situations or procedures into algebraic expressions or formulae derive an equation (or two simultaneous equations), solve the equation(s) and interpret the solution	

Notes: including the solution of geometrical problems and problems set in context.

A22

Basic foundation content	Additional foundation content	Higher content only
	solve linear inequalities in one variable	solve linear inequalities in one or two variable(s), and quadratic inequalities in one variable
	represent the solution set on a number line	represent the solution set on a number line, using set notation and on a graph

Notes: students should know the conventions of an open circle on a number line for a strict inequality and a closed circle for an included boundary. See also [N1](#)

In graphical work the convention of a dashed line for strict inequalities and a solid line for an included inequality will be required.